Arbitrating Between Pre-Activation and Surprisal as the Source of Predictability Effects in Reading

Ryan Buggy
University of Massachusetts Amherst

Stephanie Cho
Stanford University

Cory Shain
Stanford University

Adrian Staub
University of Massachusetts Amherst

Background

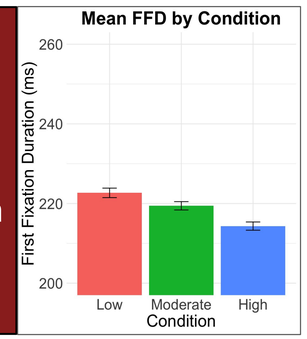
- Reading times, a proxy for processing difficulty, are negatively correlated with words' **contextual predictability**[1,2,3]
- Two competing explanations: the **pre-activation** of lexical items in memory,[4,5] or the information theoretic concept of **surprisal**[6,7,8]
- Different predictions about the form of the relationship between predictability and processing difficulty: **linear**[9,10] (pre-activation), or **logarithmic** (Surprisal)[6,7,8]
- While evidence has accumulated for a logarithmic relationship (i.e., for Surprisal),[8,11,12] **Brothers & Kuperberg**[13] ("BK21") found strong evidence for a **linear relationship** in a **self-paced reading** study with well controlled materials
- We use the **BK21 stimuli in an eye-tracking paradigm** to see if their results replicate
- We also investigate whether the source of predictability estimates—**cloze values or GPT-2-generated estimates**—affect the form[14]

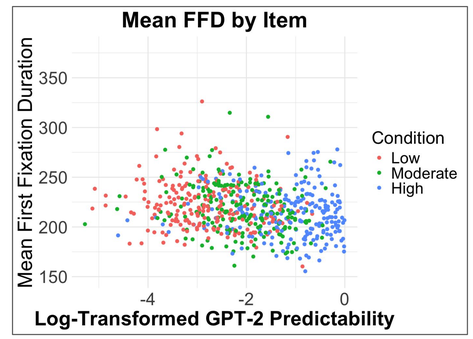
Methods

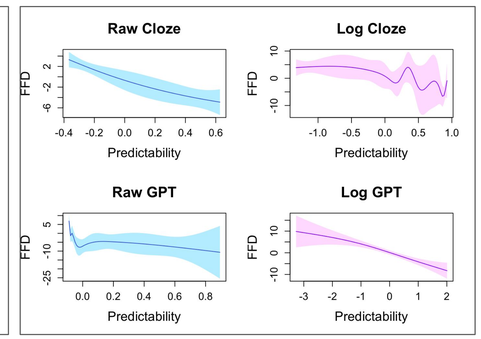
- **Eye-tracking** during reading
- **86 subjects** (66 females; median age = 19)
- Stimuli from BK21:
  - **216 sentence triplets** (low, medium, and high critical word predictability; average cloze of 0.01, 0.2, and 0.91, respectively)
  - Each subject saw only one sentence in each triplet

Low Predictability:	My picky younger brother refused to eat the *banana* at breakfast.
Moderate Predictability:	Because of fruit flies, he threw away the *banana* at breakfast.
High Predictability:	The hungry monkey was trying to peel the *banana* at breakfast.

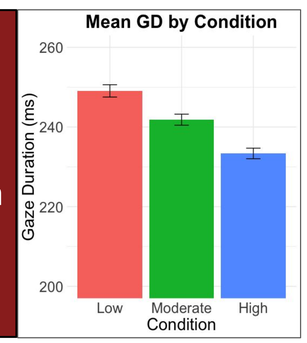
First Fixation Duration

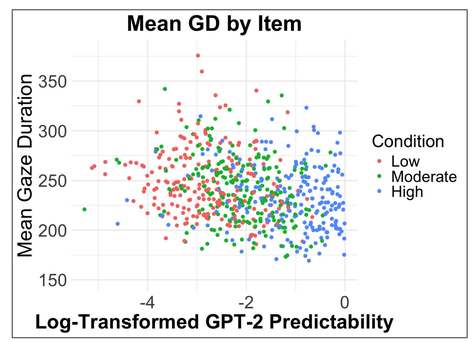


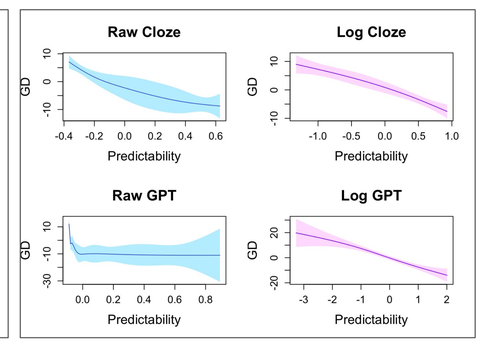




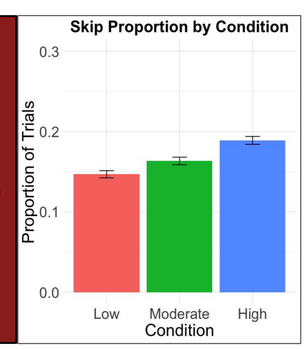
Gaze Duration

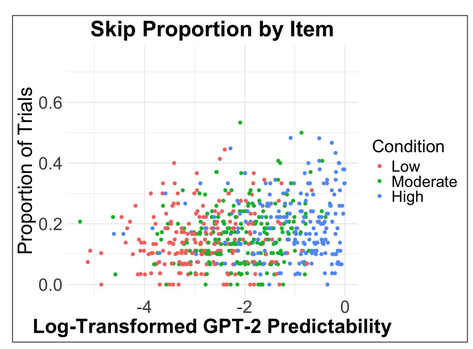


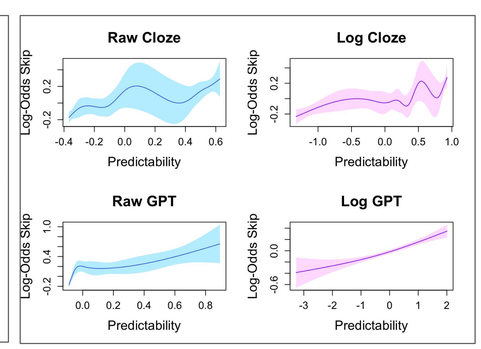




Skipping







Key LME Results

	Cloze Models: AIC (Akaike Weight)		GPT-2 Models: AIC (Akaike Weight)	
Metric	**Linear**	**Log**	**Linear**	**Log**
FFD	158,073.1 (0.000)	158,069.2 (0.002)	158,079.3 (0.000)	**158,056.6 (0.998)**
GD	165,313.3 (0.001)	165,307.9 (0.021)	165,344.2 (0.000)	**165,300.2 (0.978)**
Skip	15,401.4 (0.016)	15,401.3 (0.017)	15,406.7 (0.001)	**15,393.2 (0.966)**

Conclusion

- Clear evidence for a **logarithmic relationship**
- Persists across **all analyses** (e.g. linear and logarithmic terms, log-transformed dependent measures, log-odds of skipping an item)
- Models using **GPT-2 predictability estimates** outperform those using cloze norms
- Strong support for **Surprisal**, contrary to BK21

Preprint and References

Article Preprint

Poster References